

openPangu实验记录

<https://ai.gitcode.com/ascend-tribe/openpangu-embedded-7b-model>

step分隔符对应关系

openPangu中 `\n\n` 在tokenize后仍为其本身

python

全屏

```
1 text = "so yes, 1/3 = 4/12.\n\nAnd 5/12 is already in twelfths.\n\nSo, P(C) = 12/12 - 4/12 - 5/12\n\nNow,"
2 tokens = tokenizer.tokenize(text)
3 print(tokens)
4
5 # ['so', '_yes', '<0x2C>', '_1', '/3', '_=', '_4', '/12', '.\n\n', 'And', '_5', '/12', '_is',
   '_already', '_in', '_tw', 'elf', 'ths', '.\n\n', 'So', '<0x2C>', '_P', '(C', '<0x29>', '_=', '_12', '/12', '_-', '_4', '/12', '_-', '_5', '/12', '\n\n', 'Now', '<0x2C>']
```

token特性

将text转换为id时若add_special_tokens=True, 会在句首增加 `<s>`, 即bos_token

Chat Template

python

全屏

```
1 sys_prompt = "Please reason step by step, and put your final answer within \\boxed{}."
2 problem = "This is a dummy problem."
3 messages = [
4     [{"role": "system", "content": sys_prompt},
5     [{"role": "user", "content": problem}
6 ]
7 prompt = tokenizer.apply_chat_template(messages, tokenize=False, add_generation_prompt=True)
8 print(prompt)
9
10 # [unused9]系统: Please reason step by step, and put your final answer within \\boxed{}. [unused10]
   [unused9]用户: This is a dummy problem. [unused10] [unused9]助手:
11
```

Thinking模式切换

openPangu是没有 `<think>` 与 `</think>` token的

1. 慢思考模式: prompt (无任何设置)
2. 快思考模式: prompt + " /no_think"
3. 自动切换快慢思考模式: prompt + " /auto_think"

CoT拆分

thinking content: [unused16] [unused17]

content: [unused17] [unused10]

其中:

- [unused16]: 类似`<think>` token
- [unused17]: 类似`</think>` token
- [unused10]: eos_token_id

 发现[unused 17]并不能很好的结束thinking，后续依然会思考，并输出多个\n\n

Tokenizer特性

默认情况下（未明确指定`add_special_tokens=False`），使用tokenizer将文本直接转换为ids时均会在开头增加`<s>`（id为1）。但若先将文本转换为token，再转换为id，则不加`<s>`。

默认情况下，模型输出时，最后一个token为[unused10]（id为45892）（qwen系列也会输出`< | end_of_sentence | >`）

模型输出的第一个token总是[unused16]。而qwen系列模型的`<think>`是包含在提示词中的，并非由模型生成。

▼python ▼

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...

```
1
2 # 当return_tensors="pt"时, prompt是否加[], input_ids都为二维。当没有设置return_tensors="pt"时, prompt不加[]
  是一维, 加[]是二维
3 # pangu与qwen都适用
4 input_ids = tokenizer([prompt], return_tensors="pt").input_ids
```

Model Details

model

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SCSS

全屏

```
1 PanguEmbeddedForCausalLM(  
2   (model): PanguEmbeddedModel(  
3     (embed_tokens): Embedding(153376, 4096, padding_idx=0)  
4     (layers): ModuleList(  
5       (0-33): 34 x PanguEmbeddedDecoderLayer(  
6         (self_attn): PanguEmbeddedAttention(  
7           (q_proj): Linear(in_features=4096, out_features=4096, bias=True)  
8           (k_proj): Linear(in_features=4096, out_features=1024, bias=True)  
9           (v_proj): Linear(in_features=4096, out_features=1024, bias=True)  
10          (o_proj): Linear(in_features=4096, out_features=4096, bias=True)  
11        )  
12        (mlp): PanguEmbeddedMLP(  
13          (gate_proj): Linear(in_features=4096, out_features=12800, bias=False)  
14          (up_proj): Linear(in_features=4096, out_features=12800, bias=False)  
15          (down_proj): Linear(in_features=12800, out_features=4096, bias=False)  
16          (act_fn): SiLU()  
17        )  
18        (input_layernorm): PanguEmbeddedRMSNorm((4096,), eps=1e-05)  
19        (post_attention_layernorm): PanguEmbeddedRMSNorm((4096,), eps=1e-05)  
20      )  
21    )  
22    (norm): PanguEmbeddedRMSNorm((4096,), eps=1e-05)  
23    (rotary_emb): PanguEmbeddedRotaryEmbedding()  
24  )  
25  (lm_head): Linear(in_features=4096, out_features=153376, bias=False)  
26 )
```

model.config

▼

java

全屏

```
1 PanguEmbeddedConfig {  
2   "architectures": [  
3     "PanguEmbeddedForCausalLM"  
4   ],  
5   "attention_dropout": 0.0,  
6   "auto_map": {  
7     "AutoConfig": "configuration_openpangu_dense.PanguEmbeddedConfig",  
8     "AutoModel": "modeling_openpangu_dense.PanguEmbeddedModel",  
9     "AutoModelForCausalLM": "modeling_openpangu_dense.PanguEmbeddedForCausalLM"  
10  },  
11  "bias": true,  
12  "bos token id": 1.
```

```
--
13  "eos_token_id": 45892,
14  "hidden_act": "silu",
15  "hidden_size": 4096,
16  "initializer_range": 0.02,
17  "intermediate_size": 12800,
18  "max_position_embeddings": 32768,
19  "model_type": "PanguEmbedded",
20  "num_attention_heads": 32,
21  "num_hidden_layers": 34,
22  "num_key_value_heads": 8,
23  "pad_token_id": 0,
24  "rms_norm_eps": 1e-05,
25  "rope_theta": 16000000.0,
26  "tie_word_embeddings": false,
27  "torch_dtype": "bfloat16",
28  "transformers_version": "4.53.2",
29  "use_cache": true,
30  "vocab_size": 153376
31 }
```

Layer Analysis

Version 0

选择layer 27（从1开始第27层，代码中输入layer_id=27）

▼ yaml ▼

全屏 ...

1	 各层 PCA+Ridge 测试集指标:
2	Layer 1: R² = 0.0591 Spearman ρ = 0.2656
3	Layer 2: R² = 0.0725 Spearman ρ = 0.2927
4	Layer 3: R² = 0.0742 Spearman ρ = 0.2939
5	Layer 4: R² = 0.0676 Spearman ρ = 0.2798
6	Layer 5: R² = 0.0663 Spearman ρ = 0.2781
7	Layer 6: R² = 0.0666 Spearman ρ = 0.2769
8	Layer 7: R² = 0.0671 Spearman ρ = 0.2775
9	Layer 8: R² = 0.0658 Spearman ρ = 0.2762
10	Layer 9: R² = 0.0661 Spearman ρ = 0.2766
11	Layer 10: R² = 0.0662 Spearman ρ = 0.2754
12	Layer 11: R² = 0.0681 Spearman ρ = 0.2774
13	Layer 12: R² = 0.0686 Spearman ρ = 0.2776
14	Layer 13: R² = 0.0714 Spearman ρ = 0.2819
15	Layer 14: R² = 0.0719 Spearman ρ = 0.2844
16	Layer 15: R² = 0.0710 Spearman ρ = 0.2837
17	Layer 16: R² = 0.0742 Spearman ρ = 0.2912
18	Layer 17: R² = 0.0738 Spearman ρ = 0.2896
19	Layer 18: R² = 0.0733 Spearman ρ = 0.2890
20	Layer 19: R² = 0.0715 Spearman ρ = 0.2871

21	Layer 20:	$R^2 = 0.0720$		Spearman $\rho = 0.2873$
22	Layer 21:	$R^2 = 0.0721$		Spearman $\rho = 0.2877$
23	Layer 22:	$R^2 = 0.0736$		Spearman $\rho = 0.2890$
24	Layer 23:	$R^2 = 0.0739$		Spearman $\rho = 0.2906$
25	Layer 24:	$R^2 = 0.0770$		Spearman $\rho = 0.2960$
26	Layer 25:	$R^2 = 0.0776$		Spearman $\rho = 0.2981$
27	Layer 26:	$R^2 = 0.0790$		Spearman $\rho = 0.3003$
28	Layer 27:	$R^2 = 0.0796$		Spearman $\rho = 0.3013$
29	Layer 28:	$R^2 = 0.0787$		Spearman $\rho = 0.2997$
30	Layer 29:	$R^2 = 0.0777$		Spearman $\rho = 0.2977$
31	Layer 30:	$R^2 = 0.0782$		Spearman $\rho = 0.2998$
32	Layer 31:	$R^2 = 0.0770$		Spearman $\rho = 0.2973$
33	Layer 32:	$R^2 = 0.0796$		Spearman $\rho = 0.2998$
34	Layer 33:	$R^2 = 0.0792$		Spearman $\rho = 0.2984$
35	Layer 34:	$R^2 = 0.0668$		Spearman $\rho = 0.2771$

Version 1

逻辑优化后

选择Layer 33

▼	yaml ▼	📄 🖥️ 全屏 ⋮		
1	Layer 1:	$R^2 = 0.2476$		Spearman $\rho = 0.5375$
2	Layer 2:	$R^2 = 0.2864$		Spearman $\rho = 0.5727$
3	Layer 3:	$R^2 = 0.2990$		Spearman $\rho = 0.5851$
4	Layer 4:	$R^2 = 0.3006$		Spearman $\rho = 0.5872$
5	Layer 5:	$R^2 = 0.3135$		Spearman $\rho = 0.5958$
6	Layer 6:	$R^2 = 0.3214$		Spearman $\rho = 0.6033$
7	Layer 7:	$R^2 = 0.3272$		Spearman $\rho = 0.6101$
8	Layer 8:	$R^2 = 0.3277$		Spearman $\rho = 0.6106$
9	Layer 9:	$R^2 = 0.3370$		Spearman $\rho = 0.6180$
10	Layer 10:	$R^2 = 0.3424$		Spearman $\rho = 0.6223$
11	Layer 11:	$R^2 = 0.3466$		Spearman $\rho = 0.6259$
12	Layer 12:	$R^2 = 0.3483$		Spearman $\rho = 0.6266$
13	Layer 13:	$R^2 = 0.3523$		Spearman $\rho = 0.6301$
14	Layer 14:	$R^2 = 0.3576$		Spearman $\rho = 0.6342$
15	Layer 15:	$R^2 = 0.3661$		Spearman $\rho = 0.6422$
16	Layer 16:	$R^2 = 0.3722$		Spearman $\rho = 0.6461$
17	Layer 17:	$R^2 = 0.3807$		Spearman $\rho = 0.6527$
18	Layer 18:	$R^2 = 0.3902$		Spearman $\rho = 0.6599$
19	Layer 19:	$R^2 = 0.3973$		Spearman $\rho = 0.6655$
20	Layer 20:	$R^2 = 0.4059$		Spearman $\rho = 0.6718$
21	Layer 21:	$R^2 = 0.4078$		Spearman $\rho = 0.6735$
22	Layer 22:	$R^2 = 0.4154$		Spearman $\rho = 0.6796$
23	Layer 23:	$R^2 = 0.4252$		Spearman $\rho = 0.6859$
24	Layer 24:	$R^2 = 0.4274$		Spearman $\rho = 0.6883$
25	Layer 25:	$R^2 = 0.4303$		Spearman $\rho = 0.6904$
26	Layer 26:	$R^2 = 0.4304$		Spearman $\rho = 0.6900$

26	Layer 26: $R^2 = 0.4294$ Spearman $\rho = 0.6899$
27	Layer 27: $R^2 = 0.4274$ Spearman $\rho = 0.6877$
28	Layer 28: $R^2 = 0.4273$ Spearman $\rho = 0.6883$
29	Layer 29: $R^2 = 0.4272$ Spearman $\rho = 0.6879$
30	Layer 30: $R^2 = 0.4268$ Spearman $\rho = 0.6871$
31	Layer 31: $R^2 = 0.4269$ Spearman $\rho = 0.6885$
32	Layer 32: $R^2 = 0.4236$ Spearman $\rho = 0.6859$
33	Layer 33: $R^2 = 0.4332$ Spearman $\rho = 0.6925$
34	Layer 34: $R^2 = 0.3637$ Spearman $\rho = 0.6384$

Version 2 (test 2)

▼	yaml ▼	📄	🖥️ 全屏	⋮
1	Layer 1: $R^2 = 0.2357$ Spearman $\rho = 0.5271$			
2	Layer 2: $R^2 = 0.2793$ Spearman $\rho = 0.5659$			
3	Layer 3: $R^2 = 0.2922$ Spearman $\rho = 0.5788$			
4	Layer 4: $R^2 = 0.2906$ Spearman $\rho = 0.5787$			
5	Layer 5: $R^2 = 0.3022$ Spearman $\rho = 0.5856$			
6	Layer 6: $R^2 = 0.3131$ Spearman $\rho = 0.5960$			
7	Layer 7: $R^2 = 0.3176$ Spearman $\rho = 0.6001$			
8	Layer 8: $R^2 = 0.3204$ Spearman $\rho = 0.6023$			
9	Layer 9: $R^2 = 0.3288$ Spearman $\rho = 0.6097$			
10	Layer 10: $R^2 = 0.3331$ Spearman $\rho = 0.6136$			
11	Layer 11: $R^2 = 0.3383$ Spearman $\rho = 0.6174$			
12	Layer 12: $R^2 = 0.3415$ Spearman $\rho = 0.6196$			
13	Layer 13: $R^2 = 0.3455$ Spearman $\rho = 0.6235$			
14	Layer 14: $R^2 = 0.3529$ Spearman $\rho = 0.6291$			
15	Layer 15: $R^2 = 0.3612$ Spearman $\rho = 0.6377$			
16	Layer 16: $R^2 = 0.3678$ Spearman $\rho = 0.6425$			
17	Layer 17: $R^2 = 0.3785$ Spearman $\rho = 0.6507$			
18	Layer 18: $R^2 = 0.3897$ Spearman $\rho = 0.6597$			
19	Layer 19: $R^2 = 0.3968$ Spearman $\rho = 0.6654$			
20	Layer 20: $R^2 = 0.4086$ Spearman $\rho = 0.6738$			
21	Layer 21: $R^2 = 0.4108$ Spearman $\rho = 0.6761$			
22	Layer 22: $R^2 = 0.4177$ Spearman $\rho = 0.6817$			
23	Layer 23: $R^2 = 0.4273$ Spearman $\rho = 0.6874$			
24	Layer 24: $R^2 = 0.4279$ Spearman $\rho = 0.6886$			
25	Layer 25: $R^2 = 0.4299$ Spearman $\rho = 0.6900$			
26	Layer 26: $R^2 = 0.4306$ Spearman $\rho = 0.6901$			
27	Layer 27: $R^2 = 0.4274$ Spearman $\rho = 0.6881$			
28	Layer 28: $R^2 = 0.4264$ Spearman $\rho = 0.6880$			
29	Layer 29: $R^2 = 0.4266$ Spearman $\rho = 0.6879$			
30	Layer 30: $R^2 = 0.4281$ Spearman $\rho = 0.6882$			
31	Layer 31: $R^2 = 0.4273$ Spearman $\rho = 0.6893$			
32	Layer 32: $R^2 = 0.4276$ Spearman $\rho = 0.6898$			
33	Layer 33: $R^2 = 0.4376$ Spearman $\rho = 0.6975$			
34	Layer 34: $R^2 = 0.3678$ Spearman $\rho = 0.6437$			

Version 2 (test 3)

Layer 26 或 Layer 33

▼

yaml

全屏

1	Layer 20: $R^2 = 0.4086$ Spearman $\rho = 0.6717$
2	Layer 21: $R^2 = 0.4131$ Spearman $\rho = 0.6752$
3	Layer 22: $R^2 = 0.4199$ Spearman $\rho = 0.6806$
4	Layer 23: $R^2 = 0.4291$ Spearman $\rho = 0.6864$
5	Layer 24: $R^2 = 0.4305$ Spearman $\rho = 0.6880$
6	Layer 25: $R^2 = 0.4325$ Spearman $\rho = 0.6892$
7	Layer 26: $R^2 = 0.4327$ Spearman $\rho = 0.6898$
8	Layer 27: $R^2 = 0.4295$ Spearman $\rho = 0.6875$
9	Layer 28: $R^2 = 0.4282$ Spearman $\rho = 0.6872$
10	Layer 29: $R^2 = 0.4282$ Spearman $\rho = 0.6873$
11	Layer 30: $R^2 = 0.4296$ Spearman $\rho = 0.6875$
12	Layer 31: $R^2 = 0.4282$ Spearman $\rho = 0.6878$
13	Layer 32: $R^2 = 0.4278$ Spearman $\rho = 0.6883$
14	Layer 33: $R^2 = 0.4396$ Spearman $\rho = 0.6968$
15	Layer 34: $R^2 = 0.3716$ Spearman $\rho = 0.6443$

Hidden Analysis

Version 0

threshold: 0.75

温和距离: $\alpha_mean_S = 1.14$

激进距离: $\alpha_all_S = 5.54$

▼

python

全屏

```
1 # hidden analysis report
2 {
3     "layer_id": 27,
4     "dim": 4096,
5     "n_hit": 47345,
6     "n_non": 55509,
7     "S_l2": 18.099870681762695,
8     "hit_to_non_min": 65.78202056884766,
9     "hit_to_non_max": 184.5203857421875,
10    "non_to_hit_min": 64.67111206054688,
11    "non_to_hit_max": 180.79078674316406,
12    "jsonl_path": "/home/ma-user/work/dataset/outputs_yulin/openPangu-Embedded-7B-V1.1/Math_Math/origin_temp
```

```
13 "hidden_dir": "/home/ma-user/work/dataset/outputs_yulin/openPangu-Embedded-7B-V1.1/Math_Math/",
14 "threshold": 0.75,
15 "center_block_size": 65536,
16 "device": "npu",
17 "confidence_q25": 0.7507111579179764,
18 "confidence_q75": 0.9260406494140625,
19 "confidence_N": 125010,
20 "w_norm": 2.055134690625292,
21 "S_norm": 18.09987190130844,
22 "b": 8.807880123351854,
23 "threshold_t": -8.807880123351854,
24 "max_w_dot_h": 143.5134631382569,
25 "margin_needed": 74.11745028506603,
26 "d_all_w": 74.11745028506603,
27 "d_all_S": 100.37578836948285,
28 "alpha_all_S": 5.545662915007766,
29 "d_mean_w": 15.25405413371759,
30 "d_mean_S": 20.658262036992678,
31 "alpha_mean_S": 1.141348521670927,
32 "w_dot_uS": 1.517510803509848
33 }
```

Version 1

threshold: 0.74

温和距离: alpha_mean_S = 0.99

激进距离: alpha_all_S = 3.97

▼ json ▼

📄 全屏 ...

```
1 {
2   "layer_id": 33,
3   "dim": 4096,
4   "n_hit": 50072,
5   "n_non": 65542,
6   "S_l2": 34.81904983520508,
7   "hit_to_non_min": 111.92693328857422,
8   "hit_to_non_max": 365.236572265625,
9   "non_to_hit_min": 114.32675170898438,
10  "non_to_hit_max": 353.8177795410156,
11  "jsonl_path": "/home/ma-user/work/dataset/outputs_yulin_gy/openPangu-Embedded-7B-V1.1/Math_Math/origin_t
12  "hidden_dir": "/home/ma-user/work/dataset/outputs_yulin_gy/openPangu-Embedded-7B-V1.1/Math_Math/",
13  "threshold": 0.74,
14  "center_block_size": 65536,
15  "device": "npu",
16  "confidence_q25": 0.7398352473974228,
17  "confidence_q75": 0.9247501939535141,
18  "confidence_N": 125722
```

```
18 confidence_N : 120722,  
19 "w_norm": 1.5205217857976203,  
20 "S_norm": 34.819037596402964,  
21 "b": 9.273947276990937,  
22 "threshold_t": -9.273947276990937,  
23 "max_w_dot_h": 141.49369775399822,  
24 "margin_needed": 99.15520214121877,  
25 "d_all_w": 99.15520214121877,  
26 "d_all_S": 138.09312030926014,  
27 "alpha_all_S": 3.9660234699744286,  
28 "d_mean_w": 24.698956613311317,  
29 "d_mean_S": 34.39815474590563,  
30 "alpha_mean_S": 0.9879122778930052,  
31 "w_dot_uS": 1.0917824486347056  
32 }
```

Version 2 (test 3)

threshold: 0.74

温和距离: alpha_mean_S = 1.17

激进距离: alpha_all_S = 4.00

▼ json ▼

📄 全屏 ...

```
1 {  
2   "layer_id": 26,  
3   "dim": 4096,  
4   "n_hit": 50972,  
5   "n_non": 67138,  
6   "S_l2": 24.135784149169922,  
7   "hit_to_non_min": 61.98368835449219,  
8   "hit_to_non_max": 180.0704345703125,  
9   "non_to_hit_min": 63.942501068115234,  
10  "non_to_hit_max": 162.03945922851562,  
11  "jsonl_path": "/home/ma-user/work/dataset/outputs_yulin_gy/test3/openPangu-Embedded-7B-V1.1/Math_Math/or  
12  "hidden_dir": "/home/ma-user/work/dataset/outputs_yulin_gy/test3/openPangu-Embedded-7B-V1.1/Math_Math/",  
13  "threshold": 0.74,  
14  "center_block_size": 65536,  
15  "device": "npu",  
16  "confidence_q25": 0.7372435331344604,  
17  "confidence_q75": 0.9215247631072998,  
18  "confidence_N": 129223,  
19  "w_norm": 3.1682395717301364,  
20  "S_norm": 24.13578860473197,  
21  "b": 18.225561513727815,  
22  "threshold_t": -18.225561513727815,  
23  "max_w_dot_h": 200.80388335693794,  
24  "margin_needed": 69.13285435387304,
```

```

25     "d_all_w": 69.13285435387304,
26     "d_all_S": 96.50850072964198,
27     "alpha_all_S": 3.9985642197212856,
28     "d_mean_w": 20.14987603118412,
29     "d_mean_S": 28.128945981365852,
30     "alpha_mean_S": 1.1654454901817393,
31     "w_dot_uS": 2.269535255595321
32 }

```

Beta

threshold: 0.75

温和距离: $\alpha_{\text{mean_S}} = 0.69$

激进距离: $\alpha_{\text{all_S}} = 3.30$

▼

json ▼

📄

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⋮

```

1  {
2      "layer_id": 18,
3      "dim": 4096,
4      "n_hit": 37477,
5      "n_non": 46689,
6      "S_l2": 9.284542083740234,
7      "hit_to_non_min": 28.435041427612305,
8      "hit_to_non_max": 63.626129150390625,
9      "non_to_hit_min": 30.764385223388672,
10     "non_to_hit_max": 58.76258850097656,
11     "jsonl_path": "/home/ma-user/work/dataset/outputs_yulin_gy/beta/openPangu-Embedded-7B-V1.1/Math_Math/ori
12     "hidden_dir": "/home/ma-user/work/dataset/outputs_yulin_gy/beta/openPangu-Embedded-7B-V1.1/Math_Math/",
13     "threshold": 0.75,
14     "center_block_size": 65536,
15     "device": "npu",
16     "confidence_q25": 0.7479931116104126,
17     "confidence_q75": 0.9229621589183807,
18     "confidence_N": 124063,
19     "w_norm": 9.040904793797575,
20     "S_norm": 9.284542745305822,
21     "b": 4.956886057741642,
22     "threshold_t": -4.956886057741642,
23     "max_w_dot_h": 170.26309026706497,
24     "margin_needed": 19.38080096197841,
25     "d_all_w": 19.38080096197841,
26     "d_all_S": 30.63149480749916,
27     "alpha_all_S": 3.299192609456492,
28     "d_mean_w": 4.033750078173237,
29     "d_mean_S": 6.375370905295122,
30     "alpha_mean_S": 0.6866650388914158,
31     "w_dot_uS": 5.720255480510202

```

```
31 w_u0c_u5 : 5.720255469519202
32 }
```

TODO List

- ☐ 当前提取thinking contents时包含了初始的[unused16], 是否需排除?
- ☐ 当前低置信词表复用qwen, 是否需重新构建?
- ☐ 从gen_logprobs中取confidence时应该跳过\n\n, 即start=end+1
- ☐ 使得所有hidden state与conf中的step数量都刚好相差1可能是不现实的

python ▾

全屏 ...

```
1 # 这里我们要算conf, 就只能算\n\n和-----\n\n这两个token的conf, 没有方式能够得到-----的conf, 因为其与\n\n是一起
2 生成的
3 # test
4 b = "so:\n\nAlign:\n\n 8855\n\n+33649\n\n-----\n\nStart from right:\n\n5+9=14"
5 tokenizer.tokenize(b)
6
7 # ['so',
8 #  ':\n\n',
9 #  'Align',
10 #  ':\n\n',
11 #  '___',
12 #  '88',
13 #  '55',
14 #  '\n\n',
15 #  '<0x2B>',
16 #  '33',
17 #  '649',
18 #  '\n\n',
19 #  '-----\n\n',
20 #  'Start',
21 #  '___from',
22 #  '___right',
23 #  ':\n\n',
24 #  '<0x35>',
25 #  '<0x2B>',
26 #  '<0x39>',
27 #  '<0x3D>',
28 #  '14']
```